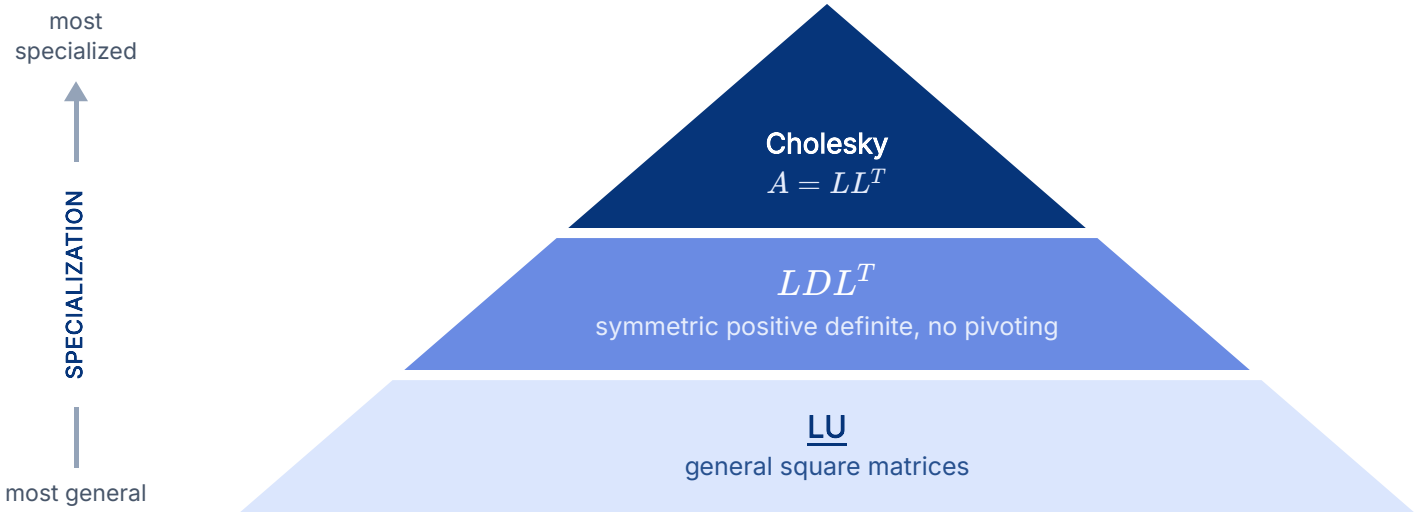
BEFORE YOU READ

## Each form specializes the one below

For a symmetric positive definite matrix, LU without pivoting always succeeds. Absorbing structure step by step leads from LU to Cholesky.




TAKEAWAY

## Why specialize

Cholesky exploits $A = A^T$ to halve the work and storage, and positive definiteness to remove pivoting. The $LDL^T$ form avoids square roots entirely.